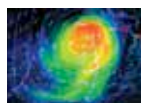


AI vs COVID19

An exploration of how AI and machine learning are being used to tackle the pandemic. <https://digit.in/jun20-98>



AI weatherman

AI could improve weather forecasts by removing error factors associated with humans. <https://digit.in/jun20-99>

FEATURE

prone to outsider attacks or breaches. The beefed up privacy is extremely invaluable in highly regulated and privacy conscious industries such as healthcare and banking.

APPLICATIONS OF TINY AI

Alongside researchers and academicians, tech giants such as Google, IBM, Amazon and Apple are also conducting research in this nascent field. Various industries and fields of technology are looking into Tiny AI to reduce computational costs, improve speed and privacy, and be more environmentally conscious. Let's take a look at Sony and Microsoft, who recently struck a deal on a Tiny AI chip to create AI-driven smart camera solutions. Sony and Microsoft are working towards embedding AI capabilities into Sony's latest imaging chip. The new AI module will feature its own processor as well as memory and will be able to analyse videos using AI tech in a self-contained system. The AI chip will be able to analyse the video footage it sees and provide metadata about what's in the frame. Since everything is stored on the device itself, privacy fears are alleviated since hackers will not be able to intercept sensitive images or videos during the transit to the servers.

The AI-powered Sony sensor is also capable of recording high-res video and conducting AI analysis simultaneously. This rapid responsiveness means that it could be used in cars to detect the driver's alertness. Since the data is not sent back to the cloud for processing, the reaction time is almost instantaneous and this technology could effectively hasten the adoption of smart-car technology.

Existing services such as voice assistants (Siri, Google Assistant, Alexa, and others), autocorrect and digital cameras will also become more efficient and faster if they adopt Tiny AI, since they wouldn't have to ping the cloud every time they needed to access the deep learning algorithms. Recently, Apple acquired Xnor.ai, a Seattle startup that specialises in low-power, edge-based AI tools. This startup developed a technology that embeds AI on the edge and

enables tasks such as facial recognition, NLP, augmented reality and other ML capabilities to be executed completely on low-powered devices rather than relying on the cloud. They achieve this by replacing the AI models' complex mathematical operations with simple, less precise binary equivalents. This technology can substantially boost the speed and efficacy of AI models and greatly reduces computational power consumption.

Yet another development of Tiny AI came with Amazon Web Services' release of the open-source AutoGluon toolkit. The AutoGluon toolkit includes a feature known as 'neural architecture search' which finds the most compact and efficient structure of a neural net to carry out a specific inferencing task. It also allows AI developers to automatically optimise the speed, accuracy and efficiency of new or existing AI models for inferencing in edge devices. AutoGluon can actually generate a high-performance AI model automatically from just a few lines of Python code. It does this by tapping into available compute resources and using reinforcement learning algorithms to find the best suited, top-performing neural network architecture for the target environment. AutoGluon can also interface with already existing AI DevOps pipelines through APIs in order to automatically alter a ML model and improve its inferencing performance.

CHALLENGES AND THE FUTURE

Research into Tiny AI is absolutely critical in order to enable it to reach its full potential. Researchers and technology firms need to efficiently man-

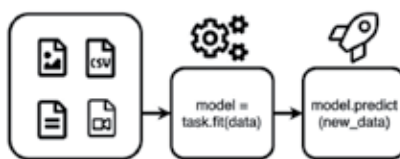
age the trade-off between shrinking the size of the AI model and maintaining accuracy and high performance for inference. For example, an attempt was made to make BERT even smaller than the Huawei version (TinyBERT), however, the accuracy was much lower than the 96 percent achieved on TinyBERT. So, researchers need to be wary of the limitations of the technology and need to also develop methods



Mieke De Ketelaere, Program Director AI at IMEC discussed Tiny AI at CES 2020

to push the envelope further. Additionally, since Tiny AI will likely push the adoption of smart-vehicles and autonomous vehicles, the room for error here is non-existent. It is vital to make Tiny AI algorithms at the edge extremely secure and ethical.

Tiny AI could potentially alter the very essence of how consumers interact with their devices and will be essential to create the future of context-aware consumer devices. It has its eyes set on an array of services and technologies and will also create new applications. Experts believe that Tiny AI is an expected evolution in the field of AI. Mieke De Ketelaere, Program Director AI, IMEC, believes that Tiny AI warrants researchers to look into new ways of running AI since firstly, the algorithms need to be run on smaller scale hardware which needs to be more power efficient, and secondly, the access to data is limited on smaller form factor devices so researchers have to come up with more novel ways to work with smaller data sets that are much more contextually aware to be as accurate as the full-fledged models. Due to these requirements and challenges, Tiny AI is not commonplace and is highly research-driven as of now, as it should be, until it is perfected. 



Amazon's AutoGluon can generate AI models with a few lines of code